

PETROPHYTUM
SYN. PETROPHYTON <i>Rock spiraea</i>
ROSACEAE
Genus of evergreen, summer-flowering subshrubs, grown for their spikes of small, fluffy flowers. Good for growing on tufa or in alpine houses. Fully hardy. Needs sun and gritty, very well-drained, alkaline soil. May be difficult to grow. Propagate by softwood or semiripe cuttings in summer or by seed in fall. In hot weather, red spider mite and aphids may be a problem. P. caespitosum . Evergreen, mat-forming subshrub. H 2in (5cm), S 1ft (30cm). Has clusters of small, spoon-shaped, silky-hairy, bluish-green leaves. Flower stems, ¾in (2cm) long, each bear a conical spike of small, fluffy, white flowers, with prominent stamens, in summer. Z5–9 P. hendersonii . Evergreen, mound-forming subshrub. H 4in (10cm), S 8in (20cm). Has branched stems covered in hairy, rounded, blue-green leaves. Conical spikes of small, cup-shaped, fluffy, white to creamy flowers are produced on 1in (2.5cm) stems in summer. Z5–8
PETRORHAGIA
CARYOPHYLLACEAE
Genus of annuals and perennials, grown for their flowers. Suitable for banks and rock gardens. Frost hardy. Prefers sun and well-drained, sandy soil. Propagate by seed in fall. Self-seeds readily. P. saxifraga , syn. <i>Tunica saxifraga</i> (Tunic flower), illus. p.361. ‘ Rosette ’ is a compact, mat-forming perennial. H 4in (10cm), S 6in (15cm). Has tufts of grasslike leaves. In summer, slender stems bear a profusion of cup-shaped, double, pale pink flowers, sometimes veined deeper pink. Z5–7
Petrosedum . See <i>Sedum</i> .
PETTERIA
PAPILIONACEAE/LEGUMINOSAE
Genus of one species of deciduous shrub, grown for its flowers. Related to <i>Laburnum</i> , differing in its erect racemes. Fully hardy. Requires full sun and fertile, well-drained soil. Propagate by softwood cuttings in summer or by seed in fall. ☉The seeds may cause stomach upset if ingested. P. ramentacea (Dalmatian laburnum). Deciduous, upright shrub. H 6ft (2m), S 3ft (1m). Mid-green leaves each have 3 ovate leaflets. Has dense, upright spikes of fragrant, laburnumlike, yellow flowers in late spring and early summer. Z7–10
PETUNIA
SOLANACEAE
Genus of annuals and perennials, always grown as annuals, with showy, colorful flowers. Half hardy. Grow in a sunny site, sheltered from wind, in fertile, well-drained soil. Deadhead regularly. Propagate by seed sown under glass in fall or mid-spring. May suffer from viruses, including cucumber mosaic and tomato spotted wilt. The many cultivars that have been produced are moderately fast-growing,

branching, bushy plants, H 6–12in (15–30cm), S 1ft (30cm), with ovate, mid- to deep green leaves, usually 2–5in (5–12cm) long. From late spring to late fall, they produce flared, trumpet-shaped, single or double flowers in a wide range of colors (available in mixtures or singly), including blue, violet, purple, red, pink, and white. Some have dark veining, central white stars, halos (throats in contrasting colors), or picotee margins.

The cultivars mostly fall into Grandiflora and Multiflora groups; other groups include Floribunda, Milliflora, and Spreading petunias:

Floribunda petunias are crosses between Grandiflora and Multiflora types. **Grandiflora** petunias have very large flowers, 3–4in (8–10cm) across, but they are easily damaged by rain and are best grown in sheltered hanging baskets and pots. **Milliflora** petunias are dwarf, arising from a genetic mutation, with flowers around 1½in (4cm) across. **Multiflora** petunias are bushier than the Grandiflora petunias, and produce smaller flowers, 2in (5cm) across, in greater quantity. They tend to be more resistant to rain damage, and are excellent for summer bedding or for a mixed border. **Spreading** petunias are low-growing, with trailing stems and flowers 2–3in (5–8cm) across. They are suitable for groundcover.

P. Double Duo Series. Compact, early-flowering, branching, Multiflora petunias. H and S 10–16in (25–40cm). Bear double, sometimes semidouble, flowers that are pink, salmon- or lavender-pink, burgundy-red, blue, or bicolored red-and-white or rose-pink and white. Z10–11 **P. Double Pirouette Series**. Compact, strongly branching, Grandiflora petunias. H and S 10–16in (25–40cm). Double flowers may be blue, burgundy-red, pink, or bicolored purple and white, red and white, or rose-pink and white. Z10–11. ‘**Orchid Mist**’ has flowers combining shades of pale pink; is weather resistant. **P. Duo Series**. Multiflora petunias. H to 1ft (30cm), S 1–3ft (30–90cm). Produce double flowers in colors including pink, lavender-pink, red, and burgundy, some with dark veining and some bicolors. Tolerate wet weather, but are best in containers or under glass. Z10–11 **P. Easy Wave Series**. Spreading, early-flowering, Multiflora petunias. H 10in (25cm), S to 2ft (60cm). Bears flowers in a wide color range that includes shell-pink, coral-orange, bright burgundy-red, and plum-purple. More mounding than Wave Series cultivars. Z10–11. ‘**Mystic Pink**’ illus. p.303. **P. Express Series**. Compact, mound-forming, Grandiflora petunias. H to 1ft (30cm), S 1–3ft (30–90cm). Trailing stems bear flowers in many hues, including pure white, red, shades of pink, and blue; some are picotees or marked with contrasting “stars.” Are weather resistant. Z10–11 **P. Frenzy Series**. Vigorous Grandiflora petunias. H 10in (25cm), S 1ft (30cm). Produce flowers in a wide color range that includes white, pink, blue, and red; flowers are sometimes veined, with “stars,” “halos,” or picotee margins.

Weather resistant. Z10–11. ‘**Blue Vein**’ illus. p.312. **P. Picobella Series**. Compact, vigorous, spreading Milliflora petunias. H 6–10in (15–25cm), S to 1ft (30cm). Produces flowers in colors including white, rose-pink, carmine-red, light lavender-blue, and blue. Weather resistant. Z10–11 **P. Shock Wave Series**. Mound-forming, spreading, semitrailing Multiflora petunias. H 8–10in (20–25cm), S 2–3ft (60–90cm). Bear prolific, single flowers in ivory, pink shades, pink with dark veins, purple, or rose. Z10–11. ‘**Denim**’ illus. p.311. **P. Sophistica Series**. Grandiflora petunias. H and S 10–15in (25–38cm). Bear pink, blue, lime-green, or dark reddish-burgundy flowers. Z10–11. Flowers of ‘**Antique Shades**’ combine pale pink and cream. ‘**Lime Bicolor**’ has irregularly splashed, lime-green and pink flowers and is good in containers. **P. ‘Stars and Stripes’**. Compact, Grandiflora petunia. H and S 10–12in (25–30cm). Produces blue, crimson, red, or rose-pink flowers, each petal marked centrally with a bold, white stripe. Z10–11 **P. Surfinia Series**. Strongly trailing petunias. H 9–16in (23–40cm), S 1–3ft (30–90cm). More vigorous and branching than many other petunias, freely bearing Grandiflora- and Multiflora-sized flowers in over 50 colors and forms, some double and some with veins. They tolerate wet weather well; are ideal for hanging baskets and window boxes. Available only as young plants, propagated by softwood cuttings. Z10–11. **SURFINIA BLUE VEIN (‘Sunsolos’)** illus. p.311. **SURFINIA LIME (‘Keiyeul’)** illus. p.316. ‘**Surfinia Purple**’ is a vigorous Grandiflora petunia, bearing abundant, magenta flowers with purple veining. ‘**Surfinia White**’ has pure white flowers with dark throats. **P. Tidal Wave Series**. Spreading, dense-growing, seed-raised Multiflora petunias. H 8–10in (20–25cm), S to 3ft (90cm). Bear purple-pink, bright pink, silver-pink, or cherry-red flowers, some with darker centers. Are weather resistant and suitable for groundcover. Z10–11 **P. Wave Series**. Floribunda petunias with trailing stems. H 4–6in (10–15cm), S 36–42in (90–110cm) or more. Single flowers may be white, pink, lavender-pink, pale lilac-blue, deep blue, and purple, sometimes with contrasting centers or darker veining. Excellent for temporary groundcover; keep well fertilized for best results. Z10–11

PHALARIS
PHAEDRANASSA
AMARYLLIDACEAE
Genus of bulbs with tubular, often pendent flowers. Frost tender. Needs full sun or partial shade and fairly rich, well-drained soil. Feed with high-potash fertilizer in summer. Reduce watering in winter. Propagate by seed or offsets in spring. P. carmiolii illus. p.410.

Phaedranthus buccinatorius. See *Distictis buccinatoria*. **Phaiophleps biflora**. See *Olsynium biflorum*.

PHAIUS
ORCHIDACEAE
See also ORCHIDS. P. tankervilleae (illus. p.467). Semi-evergreen, terrestrial orchid. H 3ft (1m). Leaves are broadly ovate, ribbed and 2ft (60cm) long. Tall spikes of flowers, 3½in (9cm) across, brown within, silvery-gray outside and each with a long, red-marked, pink lip, open in early summer. Provide partial shade in summer. Z11–12 (min. 59°F/15°C; max. 86°F/30°C)

PHALAEENOPSIS
Moth orchid
ORCHIDACEAE
See also ORCHIDS. P. Allegria gx . Evergreen, epiphytic orchid for a warm greenhouse. H 3ft (1m). Broadly ovate, fleshy leaves are 6in (15cm) long. Bears sprays of white flowers, to 5in (12cm) across; flowering season varies. Needs shade in summer. Z12 (min. 64°F/18°C; max. 86°F/30°C)

P. cornu-cervi. Evergreen, epiphytic orchid for a warm greenhouse. H 6in (15cm). Has broadly ovate leaves, 4in (10cm) long. Yellowish-green flowers, 2in (5cm) across, with brown marks, are borne successively, either singly or in pairs, in summer. Needs shade in summer. Z12 (min. 64°F/18°C; max. 86°F/30°C) **P. Lady Pink Lips gx** (illus. p.466). Evergreen, epiphytic orchid for a warm greenhouse. H 6in (15cm). Broadly ovate leaves are 4in (10cm) long. Tall, pendent spikes of pink flowers, 3½in (9cm) across, are produced at varying times of year. Requires shade in summer. Z12 (min. 64°F/18°C; max. 86°F/30°C) **P. Lundy gx** (illus. p.467). Evergreen, epiphytic orchid for a warm greenhouse. H 6in (15cm). Broadly ovate leaves are 9in (23cm) long. Has sprays of red-striped, yellow flowers, 3in (8cm) across; flowering season varies. Grow in shade in summer. Z12 (min. 64°F/18°C; max. 86°F/30°C)

PHALARIS
POACEAE/GRAMINEAE
See also GRASSES, BAMBOOS, RUSHES, and SEDGES. P. arundinacea var. picta , syn. <i>P.a. ‘Picta’</i> (Gardener’s garters). Evergreen, spreading, perennial grass. H to 1.5m (3ft) in flower, S indefinite. Has broad, white-striped leaves and bears narrow panicles of spikelets in summer. Z4–9. ‘ Feesey ’